

INDA OBANYI

AI/ML ENGINEER • AI PRODUCT BUILDER • DATA SCIENTIST • SOFTWARE DEVELOPER

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PROFESSIONAL PROFILE

AI/ML Engineer, Data Scientist, and technology builder with an **HND in Computer Science** and professional AI/ML training through **3MTT**.

I build intelligent systems that move beyond experimental notebooks into practical applications. My work spans **machine learning, predictive analytics, explainable AI, natural language processing, agentic AI, FastAPI backends, interactive applications, data analysis, and AI-powered product development**.

My portfolio includes solutions for **financial fraud detection, agricultural intelligence, flood-risk management, recruitment automation, airline customer-experience prediction, education technology, and multimodal/agentic AI**.

I am particularly interested in building AI systems that solve meaningful problems in **Africa and emerging markets**, combining technical implementation with product thinking, user experience, and real-world applicability.

TECHNICAL EXPERTISE

Artificial Intelligence & Machine Learning

Machine Learning • Predictive Modelling • Classification • Regression • NLP • Explainable AI • Feature Engineering • Model Selection • Cross-Validation • Hyperparameter Tuning • SHAP • Autonomous/Agentic AI • Decision Systems

Data Science & Analytics

Python • Pandas • NumPy • Exploratory Data Analysis • Statistical Analysis • Data Cleaning • Feature Engineering • Data Visualization • Predictive Analytics • Business Insights

Machine Learning Technologies

Scikit-learn • Random Forest • Logistic Regression • Gradient Boosting • HistGradientBoosting • TF-IDF • SHAP • Model Evaluation • ROC-AUC • Precision • Recall • F1-Score

Backend & Application Development

FastAPI • REST APIs • SQLAlchemy • Pydantic • JWT Authentication • Streamlit • SQLite • PostgreSQL • Redis • API Security

Frontend & Product Development

Next.js • TypeScript • React • Tailwind CSS • Responsive UI • Product Prototyping • Dashboard Development • UI/UX Design

Engineering & Development Tools

Git • GitHub • Docker • Alembic • VS Code • Jupyter Notebook • Anaconda • WSL

SELECTED PROJECTS

NEXUS

Intelligent Farm Decision & Autonomous Management System

AI/ML • Agentic AI • Agriculture Intelligence • Decision Systems • Full-Stack Development

Designing and developing an intelligent agricultural platform built around the decision cycle:

Sense → Understand → Predict → Reason → Decide → Recommend → Act → Observe → Learn

NEXUS is being developed as more than a farm-record application. Its long-term architecture combines agricultural data, predictive intelligence, AI reasoning, recommendations, autonomous actions, and continuous learning.

Core capabilities include:

- Farm, field, crop, and livestock intelligence
- Farm observations and evidence tracking
- Crop health monitoring
- Disease and pest-risk intelligence
- Irrigation and fertilizer recommendations
- Yield and harvest prediction

- Livestock health intelligence
- Weather and environmental risk assessment
- Market opportunity intelligence
- Farm task and recommendation management
- Autonomous AI decision-making

Built an operational local pilot with farm registration, field management, observations, evidence review, task management, history tracking, and persistent SQLite storage.

INDAVERSE

Multimodal & Agentic AI Creation Platform

FastAPI • Next.js • PostgreSQL • Redis • Docker • Agentic AI

Building a multimodal AI platform designed to transform natural-language intent into digital and creative outputs through intelligent planning, model routing, specialized agents, verification, and iterative refinement.

Engineering completed includes:

- FastAPI backend architecture
- Next.js frontend foundation
- PostgreSQL database
- Redis infrastructure
- Dockerized development services
- SQLAlchemy ORM
- Alembic database migrations
- JWT authentication
- Secure HTTP-only authentication cookies
- User registration and login
- Protected user sessions
- Project creation and management APIs

IndaVerse explores the future of **multimodal, agentic, and generative AI product development** through a scalable full-stack architecture.

FRAUDGUARD AI

Enterprise Mobile Money Fraud Detection Platform

Python • Machine Learning • FastAPI • Streamlit • SQLite

Developed an end-to-end fraud detection platform for analysing mobile-money transactions and identifying potentially fraudulent activity.

Key capabilities:

- Real-time transaction analysis
- Fraud-risk prediction
- Feature-engineered transaction intelligence
- Fraud monitoring
- Transaction database
- Alert system
- AI assistant
- FastAPI inference backend
- Streamlit frontend
- API-key authentication
- Administrative authentication

Engineered transaction features including balance changes, balance inconsistencies, transaction ratios, transaction types, and large-transaction indicators.

The project demonstrates the integration of **machine learning, backend APIs, security, persistent data storage, and user-facing AI applications.**

FLOODGUARD AI

Nigeria Flood Risk & Emergency Intelligence Platform

AI • Disaster Technology • Risk Intelligence • Product Design

Designed an AI-powered national flood-risk and emergency-response platform focused on improving access to actionable disaster information across Nigeria.

Core concepts include:

- State, LGA, and community-level flood intelligence
- National flood-risk heat map
- Emergency alerts
- AI flood-safety assistant
- Evacuation recommendations
- Emergency shelter tracking
- Resource allocation
- Historical flood information
- Emergency command dashboard
- SMS accessibility
- USSD accessibility
- WhatsApp-based assistance

FloodGuard AI was designed around accessibility, ensuring critical information could potentially reach both smartphone and non-smartphone users.

RESUME SCREENING AI

Intelligent Candidate Screening & Job-Matching System

Python • NLP • TF-IDF • Scikit-learn • Streamlit

Developed an AI-powered recruitment application capable of analysing individual or batches of resumes against target job requirements.

Features include:

- PDF and DOCX resume processing
- Single and batch screening
- TF-IDF similarity analysis
- Controlled relevance scoring
- Skills extraction
- Experience analysis
- Candidate assessment scoring
- Fit / Needs Review classification
- Strength and concern explanations
- CSV result export
- Interactive Streamlit interface

Compared multiple classification approaches including **Logistic Regression, Random Forest, and HistGradientBoosting**.

AIRLINE SATISFACTION AI

Explainable Customer Satisfaction Prediction System

Random Forest • SHAP • Python • Streamlit • Predictive Analytics

Developed an explainable machine learning application for predicting airline passenger satisfaction from a dataset of approximately **129,880 passenger records**.

The final Random Forest model achieved:

Accuracy: 95.61%

Precision: 96.93%

Recall: 94.98%

F1 Score: 95.95%

ROC-AUC: 99.28%

Integrated SHAP explainability to identify major drivers of customer satisfaction, including inflight entertainment, seat comfort, customer type, online booking experience, leg-room service, online support, class, and onboard service.

Application features:

- Satisfaction prediction
- SHAP explanations
- Action prioritization
- What-if simulation
- Passenger profiles
- Probability visualization
- Business recommendations
- Prediction history

ADDITIONAL PROJECT WORK

Marketing ROI & Multi-Channel Analysis

Applied exploratory analysis, statistical modelling, multiple regression, OLS, multicollinearity analysis, and VIF to evaluate relationships between marketing investment and business outcomes.

NBA Player Longevity Prediction

Applied data preprocessing, feature engineering, predictive modelling, evaluation, and visualization to investigate factors associated with professional basketball career longevity.

DigiBridge

Designed a digital platform concept connecting users with opportunities, resources, and pathways for digital growth.

AI-Powered Education Platform

Designed an education technology ecosystem for Primary 1–SS3 with CBT assessments, progress tracking, results, educational content, games, parent/teacher accounts, payments, and role-based access.

Legends of the Talking Drum

Developed an African cultural adventure-game concept centered on **Duniya**, a warrior princess, combining African storytelling, heritage, exploration, and interactive entertainment.

PROFESSIONAL EXPERIENCE

Mathematics Tutor

Divine Excellence Dynamic Academy | 2023 – Present

- Teach Mathematics and support learners across secondary-school levels.
- Develop structured learning materials and assessments.
- Apply technology to improve teaching, assessment, and learner engagement.
- Coordinate and support academic and project-based programmes.
- Mentor students in problem-solving, critical thinking, and academic development.
- Contribute to school educational innovation and extracurricular learning initiatives.

Secondary Scholars League (SSL)

Contribute to the planning and coordination of an interdisciplinary student-development programme incorporating:

- Mathematics competitions
 - Reading and literacy
 - Spelling competitions
 - Public speaking
 - Vocational activities
 - STEAM projects
 - Team-based academic challenges
 - Leadership development
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Computer Instructor

Honors Hall Academy | 2020 – 2022

- Taught fundamental computer and digital-literacy concepts.
 - Guided learners through practical computer applications.
 - Supported technology-based learning activities.
 - Helped students develop confidence using digital tools and computer systems.
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EDUCATION

Higher National Diploma (HND) — Computer Science

Lagos State Polytechnic
2019

PROFESSIONAL CERTIFICATION

AI/ML — 3MTT

Certified through Nigeria's **3 Million Technical Talent (3MTT)** programme in the AI/Machine Learning learning track.

Core areas of development include:

Artificial Intelligence • Machine Learning • Python • Data Science • Predictive Modelling • Model Development • Practical AI Applications

INNOVATION & PROFESSIONAL DEVELOPMENT

Airtel NextGen / 3MTT Knowledge Showcase

Developed **FloodGuard AI** as a technology-driven approach to flood preparedness and emergency intelligence.

Kaggriculture

Worked on autonomous agricultural decision-making concepts that contributed to the evolution of the broader **NEXUS** intelligent agriculture vision.

AI Nexus Collective

Contributed to an AI/ML-focused community centered on learning, building, collaboration, hackathons, career development, and practical technology innovation.

WHAT I BUILD

I am particularly interested in:

Artificial Intelligence & Machine Learning

Building predictive, explainable, and increasingly autonomous AI systems.

Agentic AI

Developing systems capable of reasoning, planning, decision-making, using tools, and learning from outcomes.

AI Products

Taking machine learning beyond notebooks into useful applications with APIs, databases, interfaces, security, and user experiences.

AI for Africa

Building technology around real challenges in agriculture, financial security, education, climate resilience, employment, and digital inclusion.

PROFESSIONAL STRENGTHS

Problem Solving • Analytical Thinking • Product Thinking • Machine Learning • AI System Design • Software Development • Research • Technical Communication • Teaching & Mentoring • Leadership • Project Development • Continuous Learning

CAREER FOCUS

I am building my career at the intersection of **Artificial Intelligence, Machine Learning, Data Science, and Software Engineering**.

I am open to opportunities including:

AI/ML Engineer • Machine Learning Engineer • Junior ML Engineer • Data Scientist • AI Engineer • Applied AI Engineer • Python Developer • AI Product Engineer • AI/ML Internship & Fellowship Opportunities

My long-term objective is to contribute to the development of intelligent systems that are not only technically capable, but also **useful, accessible, scalable, and relevant to real-world challenges**.

CONTACT

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References available upon request.